

Fall 2022

Biotechnology Program, Department of Life Sciences
University of New Hampshire, Manchester



BIOT 413.LM4:

PHAGE DISCOVERY LAB

Instructor: Dr. Kristen Johnson

Office: Room 630

Office Hours: Mondays 1-2 pm by zoom; and by appointment

Lab (Rm 560): Tu/Th, 9:30-11:30 am

COURSE DESCRIPTION

This course has been developed to implement SEA-PHAGES (Science Education Alliance-Phage Hunters Advancing Genomics and Evolutionary Science), the first of a two-semester, **discovery-based undergraduate research course** that begins with simple digging in the soil to find new bacterial viruses but progresses through a variety of microbiology techniques to purify and characterize the viruses and eventually to complex genome annotation and bioinformatic analyses. By finding and naming their own bacteriophages, students develop a sense of **project ownership** and carry out a personal research project.

COURSE OBJECTIVES: *Students will...*

- (1) Describe the biology of phage and their lifecycles.
- (2) Explain the experimental processes associated with isolating and characterizing phage.
- (3) Develop microbiology research skills to study phage.
- (4) Evaluate and analyze data generated from experiments and make hypotheses about these results.
- (5) Develop oral presentation skills and distill results into a final poster presentation.

Method of Evaluation	Point Value
Homework: Math Packets I and II (5 pts ea)	10
Online Lab Notebook and Demonstration of Benchmarks	40
2 Journal Club Article Online Annotations, Figure Annotations, Questions, and Discussion (10 pts ea)	20
3 Online Quizzes (5 pts ea)	15
Attendance, Lab Citizenship & Research Habits of Mind	10
Lab Meeting Presentations (5 pts ea)	10
Final Poster / Presentation	20

REQUIRED MATERIALS

LAB MANUAL

<https://seaphagesphagediscoveryguide.helpdocsonline.com/home>

COURSE PAGE

<https://mycourses.unh.edu/courses/97716>

Weekly Modules contain:

- All course materials, including:
 - syllabus
 - Power Point presentations
 - Handouts
 - lab content
 - useful websites

LAB ESSENTIALS

Lab Coat

Protective Eyewear

Laptop Computer

* CONTACT INFO

kristen.johnson@unh.edu
603-641-4177 (office)
Zoom for office hours:
unh.zoom.us/j/3094843975

COURSE REQUIREMENTS and ASSESSMENTS OVERVIEW:

Daily Preparation

It is expected that students will read (or view videos) of all background information, lab introductions and protocols prior to the beginning of each lab. In addition, Benchling lab notebook entries (steps 1-3) will be started for each day prior to entry into lab.

Homework: Math Packet / Online Lab Quizzes

At the start of the semester, you will work on math problems that are important to your research in this lab course. It is expected that you complete the packets on your own to optimize your learning. One lab quiz will be given during week 3 and will focus on lab math. Two lab quiz will be assigned after the midterm to assess your knowledge of phage biology and understanding of the experiments and fundamental principles behind them (dates: week of 10/11, and 11/8).

Lab Meeting Presentations

Twice during the semester, you will prepare an informal 3-5 minute class presentation of your research progress by describing your clearly labeled data on a couple of power point slides.

Lab Benchmarks / Lab Citizenship & Research Habits of Mind

Each week, I will take lab attendance. Throughout your time in lab, I will also take note of your engagement with your classmates, your completion of lab jobs (be an integral team player, not a sideline observer; there are many roles that one can take in each stage of the process), and your overall improvement in skills as the semester progresses. I will also follow your progress through your online Benchling lab notebook upkeep throughout the semester. For information regarding Benchmarks, refer to MyCourses Introductory Module.

Journal Clubs

Two phage biology research articles will be read by the class. Student assignments associated with the readings will include article annotations, figure annotations, discussion questions, and data presentation.

Final Lab Poster & Presentation: *week of 11/29*

Team project presented at the Winter Symposium through a team poster and through an in class Powerpoint Presentation in which each student must present a portion of the content.

LEARNING RESOURCES

Library: The UNH Manchester librarians are available to assist you with your research. You can contact a librarian by calling 603-641-4173 or by emailing unhm.library@unh.edu.

The following online resources provide information about library resources and services:

- **UNH Manchester Library webpage:** <https://manchester.unh.edu/library>
- **Online Research Guides:** <https://libraryguides.unh.edu/index.php?b=s>
- **Access Library Resources Remotely:** <https://libraryguides.unh.edu/remotearchive>

Reserve a study room for Zoom classes: <https://libraryguides.unh.edu/remotearchive/studyrooms>

Center for Academic Enrichment: Provides academic support services, including individual tutoring, for all students. Make an appointment through the CAE on your Canvas dashboard. Phone: (603) 641-4113. Email: unhm.cae@unh.edu

Academic Honesty:

The Biology/Biotechnology Programs at UNH Manchester will strictly adhere to the [University policy on academic honesty](#), as published in the UNH Student Rights, Rules, and Responsibilities Handbook. It is imperative that you review this policy to understand the rules and responsibilities about honest, academic integrity, plagiarism, and cheating. By turning in any piece of work in this course, you declare that you have read and understand the policy, and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as: submitting someone else's work - in whole or in part - as your own; collaborating on answers for individual homework assignments or allowing your own work to be used by another student; copying information from a web site or other text without proper documentation; buying a pre-written paper or lab report.

Your assignments submitted through Canvas will be routinely screened for plagiarism using sophisticated software. It is strongly recommended that you complete the [Tutorial on Plagiarism in order to fully understand the definition\(s\) and all forms of it](#).

Cheating is mainly concerned with copying on quizzes, exams or in lab, bringing crib notes into an exam or referring to notes or the textbook or any other source such as a programmable calculator, tablet, or cell phone during an exam. All electronic devices must be turned off and placed upside down on the table in front of you for the duration of the exam. Earbuds are not permitted.

Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

COURSE POLICIES

Attendance

Class attendance is important for your learning. Attendance in this course is mandatory. You are responsible for all course assignments and meeting all deadlines unless exceptions are agreed upon with the instructor ahead of time. If you need to miss class for a planned activity, let the instructor know ahead of time by email.

- Attendance will be noted during every lab and lecture section. If you must miss a lecture class for a valid reason (e.g., illness), please contact me (by email in advance) to let me know. Excessive absences (two or more) from the laboratory will result in a reduction in your lab final Benchmark grade (-2% *per absence*).
- It is your responsibility to get all lecture and lab material to make up what you have missed during any absence. Also, be sure to check MyCourses/Canvas for relevant Power Points, handouts, assignments, etc.

See the [Attendance and Class Requirements policy](#) in the SRRR.

If a student needs accommodation for a religious or cultural holiday/observance, that student is encouraged to make that request as early in the semester as possible.

Classroom Conduct

- *Cell phones:* Please silence your cell phone before you enter the lab classroom. For this lab, in many cases, you will want to use your phone to document your results. Thus, you may bring it into the lab, but keep it in a secure (CLEAN) location, until you are ready to take the pictures – i.e, you don't want to spill bacteria on it! Thus, keeping it in a zip lock bag is a good idea.
- *Computer Use:* You will want your laptop in the lab to document your lab progress in your online lab notebook. Be sure that you are careful to have it off to the side of your working area and that you wipe down computer surfaces prior to leaving the lab.

Lab Conduct

- *Food:* There is a strict policy of **NO** food or drinks in the laboratory.
- You must wear close-toed shoes, long pants (or long skirt), a lab coat and protective eye wear/goggles at **all times** while we are doing experiments in the lab.

Accessibility Services

The University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you must register with the Student Accessibility Services (SAS) office. The Student Accessibility Coordinator at UNHM is Jenessa Zurek. Please reach out to the SAS office via email at jenessa.zurek@unh.edu for registration information and disability related questions. Jenessa Zurek is available through phone and email Mondays and Wednesdays from 9am-2pm.

COVID Protocols

Unfortunately, COVID is still a part of our community. It is your responsibility to pay attention to messaging from the University (RAVE and Canvas and Email) if any COVID protocols change. You can always access current COVID protocols and requirements through the Health and Wellness Website:

<https://www.unh.edu/health/health-alert-covid-19>

We all value the health and safety of our Wildcat Community and respect everyone's unique health and risk tolerance. You are welcome to wear a mask in this classroom if you choose. It is your responsibility to obtain a mask before coming to class.

If you required to be in isolation or quarantine, the Dean of Students will send a letter to all your instructors. See extended absence policies below for temporary academic resources to support your continued learning in this course if you must miss significant class time.

A valid Wildcat Pass is required to be on campus and in this classroom. Your Wildcat Pass will be invalid if you are supposed to be in isolation or quarantine.

Temporary Academic Supports for Extended Absences with Letter

If you are dealing with an unexpected, extenuating circumstance that will keep you out of class or affect your performance for more than a day or two, reach out to Stephanie Kirylych, Director of Academic Advising, at stephanie.kirylych@unh.edu to request a letter be sent to all your faculty. Note: If you are asked to quarantine or isolate due to COVID-19 by Health & Wellness, a letter will automatically be sent to your faculty.

If you are required to miss significant class time (e.g., due to required isolation or quarantine), you will be provided temporary academic supports so that you can continue to make satisfactory progress in this course. Please contact your course instructor to discuss the specific types of supports that will be implemented during your absence. Possible options you may be provided in this class include:

- Class notes from a peer
- One-on-one meetings with instructor to catch-up on missed content
- Virtual office hours
- Access to recorded lectures (*if available*)
- Remote access for class (via zoom)
- Lecture slides
- Handouts or other materials that are distribute

Food Pantry

The UNH Manchester Food Pantry, located in room 437 is open Monday through Friday from 8:00am- 9:30pm. Any UNH Manchester community member can take what they need. If you have any questions, please email UNHM.Foodpantry@unh.edu

Emotional or mental health distress

In partnership with The Mental Health Center of Greater Manchester, UNH Manchester offers consultation visits on a walk-in basis and through telehealth appointments.

Services include:

- Free confidential screening & consultation with a licensed mental health therapist.
- Referrals to mental health or substance misuse treatment. And assistance in understanding how to afford additional treatment (with or without insurance!) or find free services.

You may email: unhm.wellness@unh.edu to make an appointment to meet with a counselor by clicking [here](#).

If you would like to connect to counseling services directly, you may do so by contacting The Greater Manchester Mental Health Center at (603) 668 - 4111.

Confidentiality and Mandatory Reporting of Sexual Violence or Harassment

The University of New Hampshire at Manchester and its community are committed to assuring a safe and productive educational environment for all students and for the university. Title IX makes it clear that violence, harassment, and discrimination based on sex and gender are Civil Rights offenses subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories such as race, national origin, and ability.

If you or someone you know has experienced sexual or relationship violence, and/or stalking and harassment, you can find the appropriate resources below:

Reporting On Campus:

- Title IX Deputy Intake Coordinator: Lisa Enright 603-641-4336. Lisa's office is located on the fourth floor in Room 439.
- UNH Manchester Security: 603-541-4101 or located in the second floor foyer

Reporting Off Campus:

- Manchester Police Department - 603-668-8711, 405 Valley St. Manchester, NH or your local police department.
- For emergencies dial 911.

Confidential Support Resources:

- YWCA, NH – 603-668-2299(24hour), 72 Concord St. Manchester, NH
- Sexual Harassment and Rape Prevention Program (SHARPP): 603-862-7233(24hour), 8 Ballard Street, Wolff House, Durham NH 03824
- The Mental Health Center of Greater Manchester: See contact information and hours above
- 24 Hour NH Sexual Violence Hotline: 1-800-277-5570
- 24 Hour NH Domestic Violence Hotline: 1-866-644-3574

Sexual Harassment and Rape Prevention Program (SHARPP): Provides free and confidential advocacy and direct services to survivors. (<https://www.unh.edu/sharpp>.) UNH Manchester students can also contact the YWCA of New Hampshire – 603-668-2299 (24hour), 72 Concord St. Manchester, NH.

BIOL 413.LM4: Phage Discovery Lab Course Schedule

Schedule subject to change depending upon experimental outcomes.

Date	Lab Activities	<u>Required</u> Pre-Lab Reading/Viewing	Assignments/ Quizzes
8/30, 9/1	Introduction to SEA-PHAGES Intro to lab work, aseptic technique, and lab safety (stations). Sample Collection.	Check out seaphages.org Tu: Read Section 1, 2; Th: Read Section 5-intro, 5.1	Soil samples due (9/1)
9/6, 8	Phage Isolation	Tu: Read Sections 4, 5.2, 5.3; Watch Videos: Direct Isolation, Plaque Assay Th: Read Sections 3, 5.5; Watch Video: Enrichment Protocol	Math Packet I Due
9/13, 15		Tu: Read Sections 5.4, 5.6; Watch Videos: spot test, picking a plaque Th: Read Sections 6-intro, 6.1, 6.2; Watch Videos: serial dilutions, plaque assay for purification	Math Packet II Due
9/20, 22	Phage Purification	Tu: Read Sections 6.3, 6.4, 6.5	Online Quiz 1
9/27, 29		Tu: Read Section 7-intro	Lab Meeting I
10/4, 6	Phage Amplification High Titer Lysate (Hard Deadline)	Tu: Read Sections 7.1, 7.2	Journal Club 1
10/11, 13			Online Quiz 2
10/18, 20			
10/25, 27	DNA Extraction and Restriction Analysis	Tu: Read Section 9-all	
11/1, 3		Tu: Read Section 10-all	Lab Meeting II
11/10	Microscopy	Tu: Read Section 8	Online Quiz 3
11/15, 17	DNA and Microscopy Analysis *DNA Submission DEADLINE*		
11/22	<i>Work on Poster</i>		
11/29, 12/1	<i>Finish Poster; Practice Poster Presentation</i>		Posters Due (11/29); Winter Symposium
12/6, 8	Lab Clean-Up and Phage Archival	Tu: Read Section 7.3	Journal Club 2

Note: Above course schedule is subject to change! Any changes will be communicated by the instructor to the students by email as soon as possible and a new syllabus issued.